IPx Projects 2024 Final Project Options

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June 17, 2024

1. Fun Physics: Exploring Special Relativity

In this project, you will learn more about a Special Relativity. This will be a super abridged an incomplete introduction, so it will still be very important to take a class, but hopefully it should be fun. You may use any sources you like to understand the problem, but your answers should only use the information given in the problem, the writeup should be entirely your own, and you should be sure to cite any sources outside this sheet that you use. If you get the right answer, but there is no explanation, or the explanation does not make sense, you will be penalized.

To begin, we'll introduce the concept of **reference frames**. Roughly speaking, these are the descriptions of the world from the perspective of a person (aka an observer), who considers themself to be at rest. So if I am in a boat traveling towards my friend on the shore, in my reference frame, he *and* the shore are both actually moving towards me! Essentially, to translate between two observers reference frames is simply to translate between different conventions of perceiving or describing the world. A natural question you might ask is this: *when is one convention better than another?* After all, if I see both my friend and the shoreline coming closer to me, I should reasonably deduce that I am actually the one moving. This might seem reasonable due to experience, but according to physics, there's nothing to say the whole shoreline wouldn't be moving. The real reason you could likely tell you were moving would be the ripples in the water behind you or the rocking of the boat. If these were not there, you would hope that the laws of physics would not care if you were moving or still.

The above idea is called the **principle of relativity**. It says that there is no "preferred" inertial (non-accelerating) reference frame. So as long as I move at a constant velocity, the physics I see should be the same physics you see.[1] But that doesn't change the fact that the distances and times in one reference frame will be different from those in another. We need a way to consistently translate between them. The most familiar and intuitive transformation

[1]For an example of a non-inertial reference frame, think of spinning around on a merry-go-round. As long as you stay at a fixed radius, you'll feel a centrifugal force pushing outwards, like an extra gravitational force. But someone outside the merry-go-round, this extra, ever-present force isn't present, so according to special relativity, you experience different physics.

between reference frame is that of **Galilean Transformations**. The reference frame of someone moving forward at speed v in the x-direction is simply

$$\begin{aligned} t' &= t \\ x' &= x - vt \end{aligned} \tag{1.1}$$

These are really easy to understand. Essentially, they just say that if I want to describe things in the reference frame of someone moving at a velocity, v, in front of me, I can just use my own description and subtract off the rate at which this reference frame is moving forward, vt, from any positions I measure. What's important to notice in this transformation is that we held *time* constant between reference frames. This is a great approximation of our daily experience, but in reality, isn't quite right.

If the speed of light is constant, regardless of reference frame, this messes up our Galilean transformations. That speed, though, is so fast, it's not surprising we missed it for so long:

$$c = 299,792,458 m/s \tag{1.2}$$

It takes a bit of work to show how one can construct transformations that respect a constant speed of light instead of a constant rate of the passage of time. We will not review that derivation here.[2] Instead, we'll go over one of the interesting effects, and quote the form of the new transformations.

To get a feel for this, we can look at what happens to time in reference frames moving with respect to each other. If we consider a "light clock," where light is bouncing between two mirrors, we notice that it will look very different if we are moving with respect to the mirrors. In frame (a), it travels up an down a distance, h, while in frame (b) it also travels across a distance vt.[3]

In frame (a), the light returns to its original position after a time $\Delta t = \frac{2h}{c}$. But in frame (b), this time is a little more complicated:

$$\Delta t' = \frac{2}{c}\sqrt{h^2 + (v\Delta t')^2}, \quad \Rightarrow \quad \Delta t' = \frac{2h/c}{\sqrt{1 - \frac{v^2}{c^2}}} \tag{1.3}$$

But if you remember what the time for this period was in the original frame ($\Delta t = \frac{2h}{c}$), this can be re-written to give us the famous time-dilation formula:

$$\Delta t' = \gamma_v \Delta t = \frac{\Delta t}{\sqrt{1 - \frac{v^2}{c^2}}} \tag{1.4}$$

Here, we introduce the *lorentz factor*: $\gamma_v = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$. It appears so often in relativity that it gets its own symbol. Notice that $\gamma \geq 1$, so that time intervals in a moving frame are always *bigger* than they are in a stationary one.

[2] For a good review, check out [Mor17,]

[3] It is important that the time "t" is measured in the moving frame where the mirrors are stationary. Can you see why?

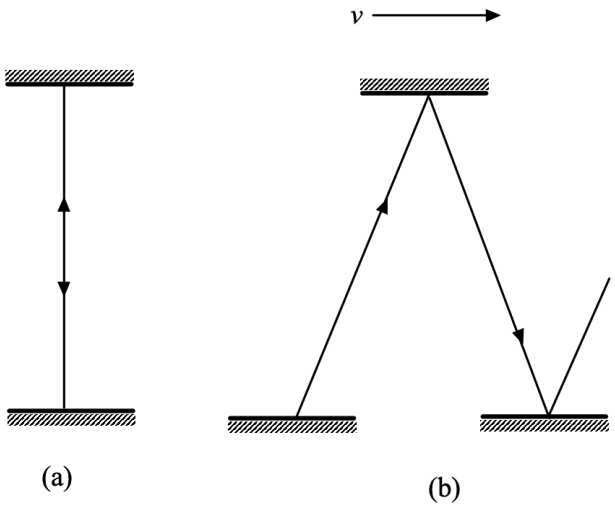


Figure 1.1: A light clock in a relatively stationary reference frame, and one moving ar a relative velocity "v"

To encode all of these changes, we define ***Lorentz Transformations***:

$$t' = \gamma_v(t - \frac{vx}{c^2}) \tag{1.5}$$

$$x' = \gamma_v(x - vt), \tag{1.6}$$

where we used the Lorentz factor, γ_v again. This makes it very easy to describe effects like time dilation, where $\Delta x = 0$, so that the time passing in a new reference frame moving at speed, v, is $\Delta t' = \gamma_v \Delta t$.

Before moving on, we review one more important concept. Since neither spatial distances or times remain constant in different reference frames, it would be nice to have some measure of "distance" between two events that doesn't depend on the reference frame. We define this "distance" as the ***Invariant Interval***, τ:

$$c\Delta\tau = c\Delta t - \Delta x, \tag{1.7}$$

where Δt and Δx are the time and spacial intervals between two events in a particular reference frame or coordinate system. You can check by hand that any Lorentz transformation of the coordinates will have the same τ.[4] Note that this interval corresponds to the time that elapses between two events that happen at the same position ($\Delta x = 0 \quad \Rightarrow \quad \Delta\tau = \Delta t$).

[4]This is actually the unique way to define a metric, or a measure of "distances" in relativity. There are lots of fancy arguments for why this is the case

Project - predict and explain the time dilation between a stationary watch and one in three related systems (i.e. find C_{td} for $\Delta t = C_{td}\Delta t_0$):

1. A watch thrown forward at velocity, u, as seen within a train moving at velocity, V, in the same direction.
2. A watch on a rocket accelerating uniformly at rate, a.
3. A watch held 10m off the ground, where the gravitational force is given by $F_g = -mg$ downwards.

Make sure to give some physical intuition for the results you obtain.

Hints:

1) Use the Lorentz transformations twice, and be careful about Δx in each frame. If you want to check your results, look up the *velocity addition formula* and try using that.
2) Write the invariant interval in differential form and check what the instantaneous velocity is for every infinitesimal increment of time, dt. What happens when $a\Delta t = c$?
3) Although nothing seems to be moving in this problem, you do have an acceleration. Try using the conservation of energy to relate this to a velocity in a different reference frame.

2. Math Tricks: Lagrangian Mechanics

In this project, you will learn more about a different way to describe classical mechanics that uses some pretty fun mathematical tools. You may use any sources you like, but make sure the writeup is your own and be sure to cite any sources that you use. If you get the right answer, but there is no explanation, or the explanation does not make sense, you will be penalized.

To start off, we'll define something called a ***Lagrangian***:

$$\mathcal{L} = K - V \tag{2.1}$$

This seems like a funny thing to use. It looks like the total energy of a system, $E = K+V$, but why would you subtract off the potential energy? Well, it turns out to be super useful if you want to predict the equations of motion. Energy is useful because it is conserved over time. So the derivative over time in a closed system is zero: $\frac{dE}{dt} = 0$. This makes it clear when one state can transition into another (they have to have the same total energy). But there are a ton of different configurations that give the same energy. Normally, to solve this issue, we would check for all of the forces acting on a particle, then use Newton's third law alongside an initial position and velocity to solve for the particle's trajectory.

But we can do things another way. Instead of choosing an initial position and velocity, we fix an initial and final position, $\vec{r}(0)$ and $\vec{r}(T)$. Then we describe something called the *action*, S, which is the time integral of this Lagrangian in terms of $\vec{r}(t)$ and the velocity, $\dot{\vec{r}}(t)$:

$$S = \int_0^T \mathcal{L}(\vec{r}(t), \dot{\vec{r}}(t))dt = \int_0^T \Big(K(\vec{r}(t), \dot{\vec{r}}(t)) - V(\vec{r}(t))\Big)dt. \tag{2.2}$$

Note that the kinetic energy is a function of the position and velocity, while the potential energy is only a function of the position.

Now, the action is not constant over time like the energy. In fact, it is an integral over time. But it is minimized for the trajectory $\vec{r}(t)$ that corresponds to the correct equations of motion. This minimization is done using something called the *calculus of variations* - calculus on functions, not variables. We won't go into details here, but it is worth checking out if you have time[5] For our purposes, all we need to know is that the equations of motion for a specific degree of freedom, $x(t)$ can be found by using the *Euler-Lagrange Equations*:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right). \tag{2.3}$$

This can be a bit tricky to use at first, since it involves both partial derivatives (using ∂, which only affect the explicit realizations of the variable in question) and absolute derivatives (using d, which addresses any dependence on a variable). Here are two quick examples: a) a ball of mass, m, falling from height h, with gravitational potential energy given by $V_g = -mgy$. and b) a ball of mass, m, connected to a spring of spring constant, k, in two dimensions. The potential energy for this spring is given by $V_s = \frac{1}{2}k|\vec{r}^2|$.

a) The Falling Ball - Since the only motion in this problem should be vertical, we will only deal with the y-coordinate. The kinetic energy is $K = \frac{1}{2}m\dot{y}(t)^2$. And the potential energy is $V_g = mgy$, so we have a Lagrangian given by

$$\mathcal{L}_g = \frac{1}{2}m\dot{y}(t)^2 - mgy. \tag{2.4}$$

Now, employing the Euler-Lagrange Equations, we have

$$\frac{\partial \mathcal{L}}{\partial y} = -mg = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}}\right) = \frac{d}{dt}(m\dot{y}(t)) = m\ddot{y}(t). \tag{2.5}$$

And we get the pretty simple and clearly correct solution that the acceleration is exactly g downwards: $\ddot{y} = -g$.

[5]For a good introduction, check out [Mor08].

b) The 2D spring - Since there are two directions in this problem, we will use both the x- and y-coordinates. The total kinetic energy is $K = \frac{1}{2}m(\dot{x}(t)^2 + \dot{y}(t)^2)$. And the potential energy is $V_s = \frac{1}{2}k|\vec{r}^2| = \frac{k}{2}(x^2 + y^2)$, so we have a Lagrangian given by

$$\mathcal{L}_s = \frac{m}{2}(\dot{x}(t)^2 + \dot{y}(t)^2) - \frac{k}{2}(x^2 + y^2). \tag{2.6}$$

Now, we employ the Euler-Lagrange Equations for $x(t)$ first:

$$\frac{\partial \mathcal{L}}{\partial x} = -kx = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = \frac{d}{dt}(m\dot{x}(t)) = m\ddot{x}(t). \tag{2.7}$$

And we get another pretty simple solution that amounts to the coordinates for the force law for a spring: $F_x = m\ddot{x} = -kx$, $F_y = m\ddot{y} = -ky$. In principle, we can solve these and more complicated differential equations with clever integration or the help of computers, but for this project, it will suffice to obtain the appropriate differential equations.

Project - predict the equations of motion for four related systems:

1. The pendulum from Project 3, of length, l for a ball of mass, m.
2. The same pendulum, but where the ball is now attached by a spring with spring constant, k.
3. A ball attached to two springs of lengths, l_1 and l_2, that are fixed at points separated horizontally by distance, L, and form angles $theta_1$ and $theta_2$ respectively (see diagram).
4. A double pendulum. One arm is length, l_1 and angled off the vertical by θ_1. At the end of this arm extends another of length l_2 angled off the vertical by θ_2.

Why or when might the Lagrangian formulation be useful from what you have seen?

Hints:

1) Be careful with signs, and try to write the kinetic energy and potential energy as functions of θ and its time derivatives (you may have to do some coordinate transformations)
2) Try to do this in radial coordinates and check your results by writing a force diagram for the problem.
3) Make sure you only solve for independent degrees of freedom. This may be easier if you switch coordinates. Again, check your work with a Force diagram. Can you see the special initial point where the ball would be stationary.
4) Be careful writing down the potential energy and with the derivatives in the Euler Lagrange equation. What happens when you set $l_2 = 0$? What about $\theta_1 = \theta_2$?

3. Coding Stable Orbits

In this project, you will learn how to code simulations of physical systems. You will be given code that simulates motion for N particles, and asked to modify it to produce three different orbits. Be sure to share both your code and the animation produced.

Newton's law of gravitation gives a gravitational potential energy of $V_G = -\frac{GMm}{r^2}$, for masses, M and m. If no other matter interferes, the system will have a constant energy given by

$$E = \frac{1}{2}M|\vec{V}|^2 + \frac{1}{2}m|\vec{v}|^2 - \frac{GMm}{|\vec{r}|}, \tag{3.1}$$

where $\vec{V}$ and $\vec{v}$ are the velocities of the masses M and m respectively, and $\vec{r}$ is the displacement between the two. We could try looking for solutions of constant energy, but it can be quite hard to avoid issues of numerical precision messing with the orbits we obtain, and some of these solutions may involve one mass escaping off to infinity.

If we want to look for particular dynamical solutions, we could start with initial positions and velocities, then use the force equations to determine future trajectories. However, we are looking for very particular trajectories: closed orbits. So we'll try something different. we know that the orbits described should return to the same position after some period, T_0. So we are going to guess a potentially unphysical periodic trajectory of the same rough shape, then slowly modify the trajectory till it obeys the right laws. We will define an initial orbit $\vec{r}^0(t) = (x^0(t), y^0(t))$.

To be a good, physical orbit, we want Newton's third law to hold:

$$\vec{F} = m\vec{a} = \vec{F}_g = -\frac{GMm}{|\vec{r}|^2}. \tag{3.2}$$

Framed in another way, whenever the quantity $d\vec{L} = m\vec{a} - \vec{F}_g \neq 0$, we have a bad orbit. But if we can minimize this quantity to zero, we're in good shape, and do not need to change the orbit. Phrased in another way, if this quantity is large, we had better adjust our orbit.

Numerically, we must break up our orbit into N discrete timesteps, Δt. We label the different times along our trajectory as follows: $\vec{r}^0(n\Delta t) = \vec{r}^0_n = (x^0_n, y^0_n)$. Given this, one condition for our orbits is that $\vec{r}^0_0 = \vec{r}^0_N$. Now, we can define a recursive algorithm to adjust the orbit at timestep, n, based on the deviation form obeying Newton's third law (when $d\vec{L} = 0$):

$$x^{p+1}_n = x^p_n - \delta(d\vec{L}). \tag{3.3}$$

This means we define the new position at timestep, n, by taking the old one and subtracting off the deviation from this balanced force law expression multiplied by some small increment, δ, that helps us approach $d\vec{L} = 0$ slowly.

Project - simulate the orbits four related trajectories:

1. A two-body circular orbit with identical masses.
2. A two-body circular orbit with different masses.
3. A three-body circular orbit with identical masses
4. A three-body figure-eight orbit with identical masses

Before starting on this project, you must confirm that you have a working version of MatLab or a similar software that you are comfortable using.

Hints:

1) Try to solve this analytically first, then define the orbits in x and y coordinates as a function of time. Test your code by setting the initial velocities slightly off the values needed for the perfectly circular orbit.
2) Try to solve this analytically first, then choose values you know will make a stable orbit. Are you tracking one orbit or two? What happens when $m/M \to 0$? Test your code by setting the initial velocities slightly off the values needed for the perfectly circular orbit?
3) Try to solve this analytically first. Are you tracking one orbit or more? Either way, make sure you include all of the potential energy terms and avoid double counting. Test your code by setting the initial velocities slightly off the values needed for the perfectly circular orbit?
4) Are you tracking one orbit or more? Can you write down an orbit(s) that look(s) qualitatively right? Make sure your initial velocities are small enough to settle into a stable orbit.

Collaboration Policy

You are allowed to collaborate with others in understanding your project, but the final writeup should be your own, individual work.

References

[] *Exploring Black Holes: Introduction to General Relativity by Taylor, Edwin F. and John Archibald Wheeler: Fine Hardcover (2000) | HHFoodBank*. en.

[Mor08] References. *Introduction to Classical Mechanics: With Problems and Solutions*. Ed. by D. Morin. Cambridge: Cambridge University Press, 2008, pp. 713–715.

[Mor17] D. J. Morin. *Special relativity for the enthusiastic beginner*. eng. OCLC: 971027525. 2017.